

Class 12: Intro to Genome Informatics

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Section 1. Proportion of G/G in a population

Downloaded a CSV file from Ensemble < https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39894595-39895595;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read this CSV file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..	Male.	Female.	Unknown.	Genotype..	forward.	strand.	Population.s.	Father
1					NA19648	(F)		A A ALL, AMR, MXL	-
2					NA19649	(M)		G G ALL, AMR, MXL	-
3					NA19651	(F)		A A ALL, AMR, MXL	-
4					NA19652	(M)		G G ALL, AMR, MXL	-
5					NA19654	(F)		G G ALL, AMR, MXL	-
6					NA19655	(M)		A G ALL, AMR, MXL	-
	Mother								
1		-							
2		-							
3		-							
4		-							
5		-							
6		-							

```
table(mx1$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G
 22  21  12   9
```

```
table(mx1$Genotype..forward.strand.) / nrow(mx1) * 100
```

```
      A|A      A|G      G|A      G|G
34.3750 32.8125 18.7500 14.0625
```

Now let's look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of G|G

```
table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100
```

```
      A|A      A|G      G|A      G|G
25.27473 18.68132 26.37363 29.67033
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Lets now dig into this further.

Section 4. Population Scale Analysis (Homework)

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

Q13: Read file “rs8067378...” into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

Sample size for each genotype:

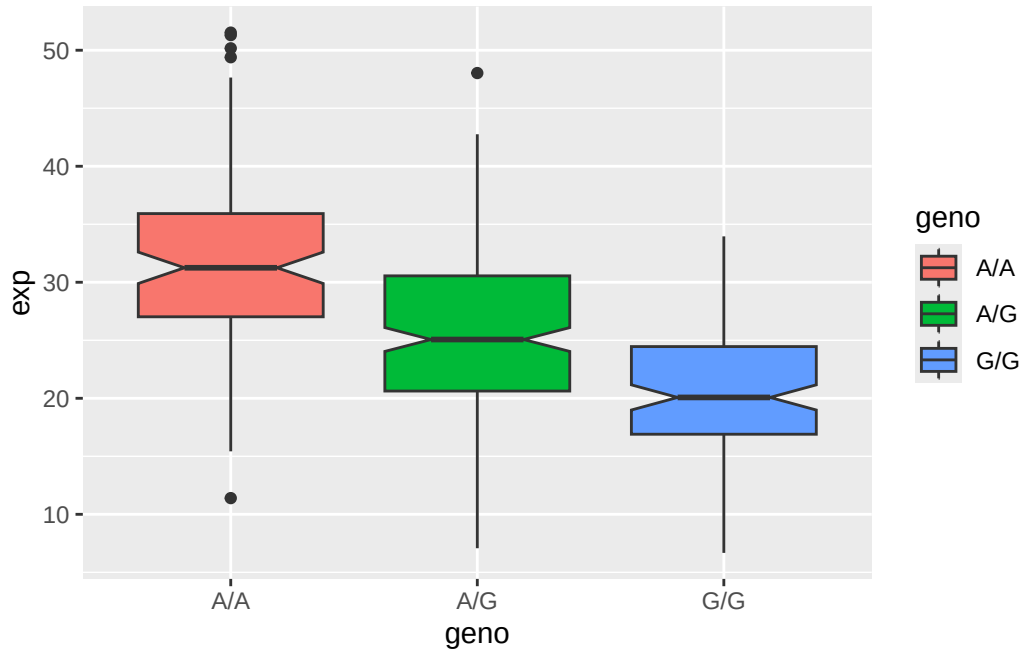
```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
library(ggplot2)
```

Let's make a boxplot to be able to find the corresponding median expression levels for each genotype:

```
ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```



Based on the generated boxplot we can infer that the median expression levels for the A/A, A/G, and G/G genotype are 32, 25, and 20 respectively.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

From the relative expression values of A/A and G/G displayed in the plot it can be inferred that the A/A genotype has higher expression level and therefore more gene activity compared to the G/G genotype. So, we can conclude that SNP does affect the expression of ORMDL3.